PINNs for SDE Option Pricing and Hedging

Quantix

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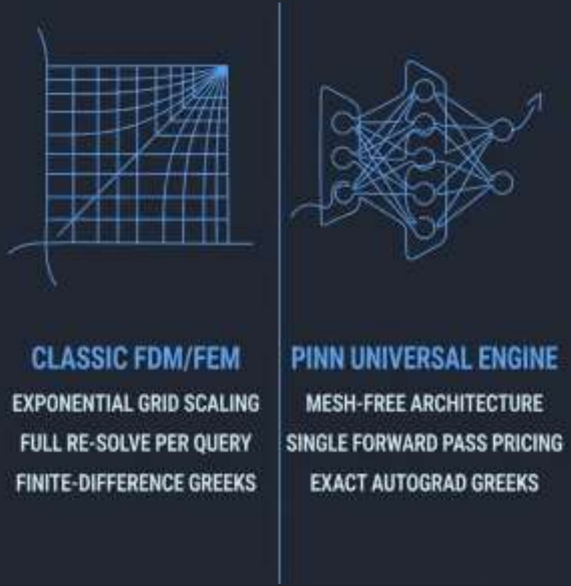
STRATEGIC IMPERATIVE

The Curse of Dimensionality: Why Classical Solvers Must Go

Finite Difference (FDM) and Finite Element (FEM) methods discretise the Black-Scholes PDE over a mesh. For a basket of **d** assets, grid complexity scales as **100d** — computationally infeasible beyond d > 3. Every parameter change demands a full re-solve.



Grid Explosion

FDM/FEM scale exponentially with asset dimensions. Multi-asset baskets are intractable.

Re-Solve Overhead

Each new (S, σ, t) query requires a complete PDE re-computation. No amortisation.

Greeks via Finite Diff

Sensitivities require additional solves per Greek — multiplicative cost at scale.

THEORETICAL FOUNDATION

Theoretical Guarantee: De Ryck & Mishra (2022)

The Black-Scholes equation is a special case of the Kolmogorov PDE class. De Ryck & Mishra (2022) provide three affirmative guarantees for PINNs on this class:

Q1 — Residual Smallness

A network exists whose PINN generalisation error can be made arbitrarily small.

Q2 — Total Error Bound

Small generalisation error guarantees small total L^2-approximation error.

Q3 — Training→Gen Gap

With N random collocation points, generalisation error is bounded above by training error + $C \cdot \ln(4N\varepsilon^6)/N$.

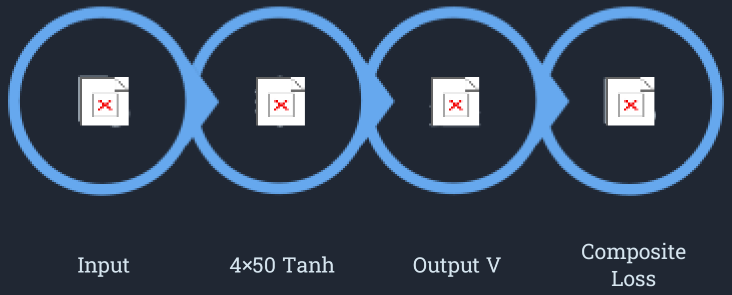
Total Error Bound (Theorem 4):
$\|\hat{u} - u\|^2_{L^2} \leq C_1|r_{PDE}|^2 + C_2|r_{BC}|^2 + C_3|r_{IC}|^2$

Key Result: Network size grows only **polynomially** in dimension d when μ and σ are constant — directly overcoming the curse of dimensionality via Dynkin's Formula and Hoeffding-type concentration inequalities.

Dynkin's Formula (SDE Connection):

$$u(s,x) = \mathrm{E}[\varphi(X_T) \mid X_s = x] - \mathrm{E}\left[\int_s^T \left(\frac{\partial u}{\partial t} + \mu \; \nabla_x u + \frac{1}{2}\mathrm{Tr}(\sigma \; \sigma \, \mathrm{Hess}_x u)\right) dt\right]$$

PINN Architecture & Composite Loss Function



Architecture: 3-input → 4×50 Tanh layers → scalar V. Xavier initialisation. ~13,051 trainable parameters. Tanh is mandatory — C² differentiability required for V_SS via autograd.

Composite PINN Loss (Black-Scholes):

$$L = \underbrace{\frac{1}{N}\sum_{i}\left|\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV\right|^2}_{\text{PDE Residual}} + \underbrace{\mathrm{MSE}(V(S,T,\sigma),\ \max(S-K,0))}_{\text{Terminal Payoff}} + \underbrace{\mathrm{MSE}(V(0,t,\sigma),\ 0)}_{\text{BC: } S=0} + \underbrace{\mathrm{MSE}(V(S_{\max},t,\sigma),\ S_{\max}-Ke^{-r(T-t)})}_{\text{BC: Deep ITM}}$$

Two-Stage Optimisation: Adam (2,500 epochs, α=0.001) for rapid basin convergence → L-BFGS (500 iterations, tol=1e-9) for super-linear fine-tuning. No labelled option prices required — physics provides the supervisory signal.

PRODUCTION VALIDATION

Real-Market Training Pipeline: 50 NSE Stocks, 16 Years

Data Scope

Daily returns for 50 NSE stocks, 2008–2024. ~197K valid (S, σ) collocation pairs from the true joint market distribution.

Volatility Construction

Rolling 252-day annualised $\sigma = \text{std}(r) \times \sqrt{252}$. Median $\sigma = 0.298$ (29.8%). Spans p10=0.215 to p90=0.492.

Price Normalisation

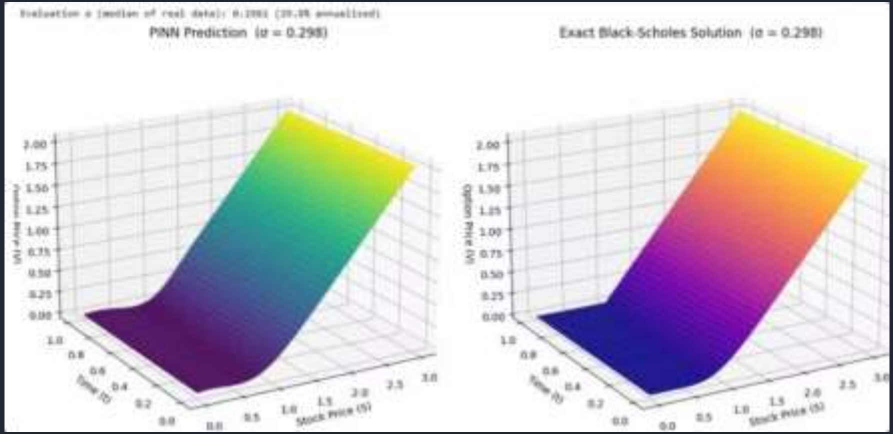
Paths reconstructed via $P_t = P_0 \times \prod(1+r_i)$, scaled to [0, S_max] using 99th percentile as robust maximum.

Regime Coverage

Training distribution spans GFC 2008, COVID 2020, and post-reform bull runs — full Indian equity market cycle.

VISUAL PROOF I

3D Price Surface: PINN vs. Exact Black-Scholes



Evaluated at median NSE volatility **σ = 0.298** across $S \in [0, 3]$ and $t \in [0, 1]$.

S = 0 (Bankrupt)

V correctly collapses to zero — boundary condition satisfied without explicit enforcement in data.

Deep ITM (S = 3)

V asymptotes to intrinsic value S − K. Cusp at K = 1.0 correctly reproduced.

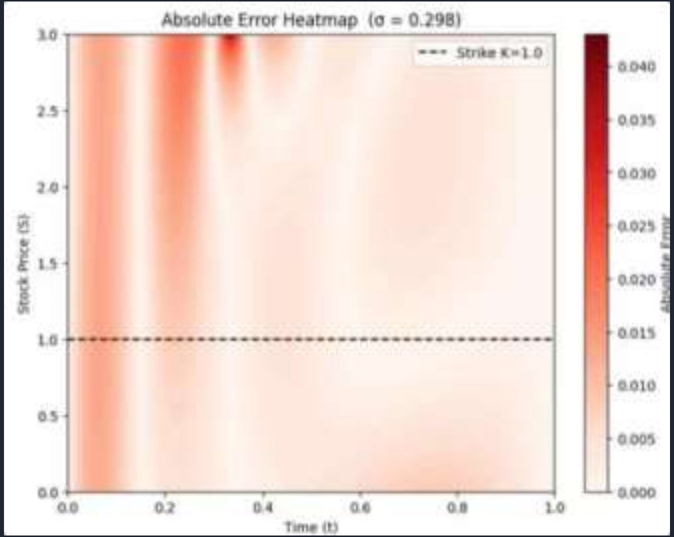
Global Structure

Colour gradients, surface curvature, and price levels are near-identical between PINN and exact B-S. Difference is palette, not values.

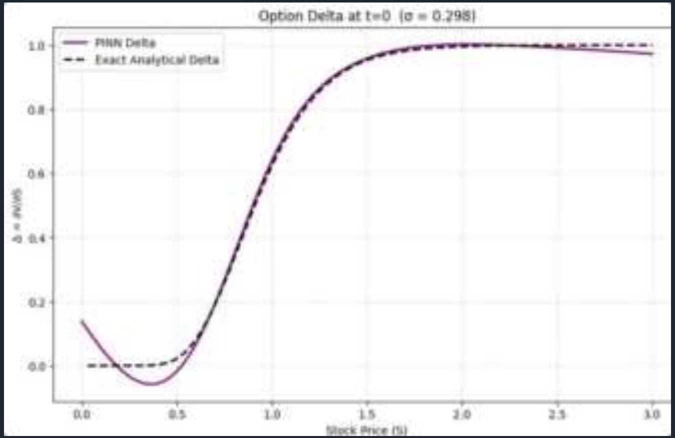
PINN learned the full B-S solution surface from physics constraints alone — zero labelled option prices used in training.

VISUAL PROOF II

Absolute Error Heatmap & The Autograd Greeks Advantage



Max error ~0.042 at early-time deep-ITM (t≈0.05, S≈3.0). Average error well below 0.010 across the domain. Near-expiry error collapses as terminal condition dominates.

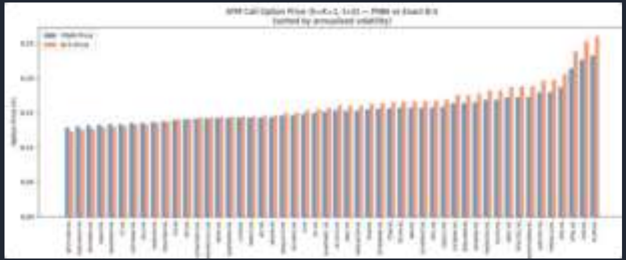


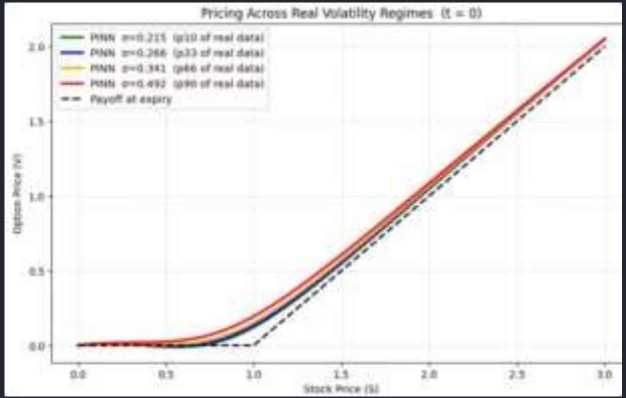
Autograd Greeks — Zero Marginal Cost:
V_t = autograd(V, t) → Theta
V_S = autograd(V, S) → Delta
V_SS = autograd(V_S, S) → Gamma
No finite-difference re-solves. Exact to machine precision. Single forward pass.

Delta agreement is excellent for S > 0.5 — covering all economically relevant hedging scenarios. Minor boundary artifact at S→ 0 due to sparse real-market data in that regime.

EMPIRICAL RESULTS

ATM Pricing Across All 50 NSE Stocks & Volatility Regimes





ATM Call (S=K=1, t=0) — Key Findings:

80th percentile of NSE stocks (σ ≤ 0.45) achieve sub-5% absolute pricing error — within practical trading tolerances. Directional ordering across all 50 stocks is perfectly preserved.

Universal engine: four volatility regimes (p10 to p90) priced in a **single forward pass** — no retraining, no re-solving.

Low-Vol Blue-Chips (<5% error)

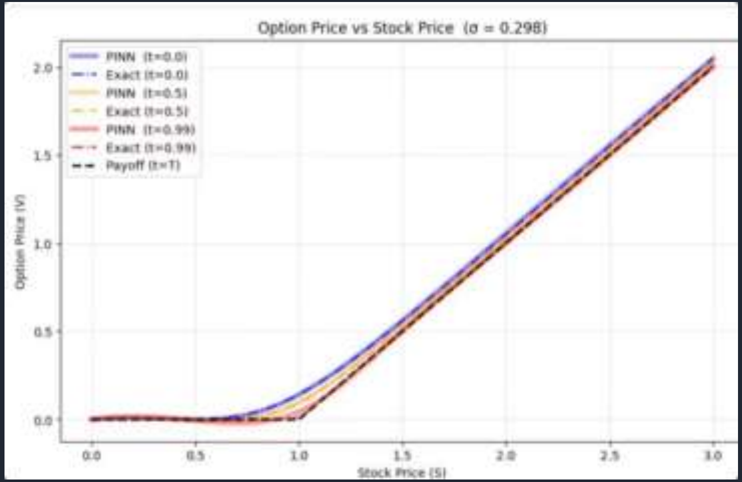
NESTLEIND (σ=0.248): PINN=0.129 vs B-S=0.123 → 4.9%
ASIANPAINT (σ=0.266): PINN=0.133 vs B-S=0.130 → 2.3%
HINDUNILVR: error ≈ 0.8%

High-Vol Tail (8–12% error)

RCOM (σ=0.614): PINN=0.233 vs B-S=0.263 → 11.4%
Expected: p90 volatility stocks are rarest in training distribution.

TIME-SLICE VALIDATION

Time-Slice Convergence & Conclusions



PINN (solid) vs. Exact B-S (dash-dot) at t=0.0, 0.5, 0.99. Lines are essentially indistinguishable at plot scale. Time decay of option value emerges purely from physics-informed loss.

Summary of Validated Capabilities

Universal Pricing Engine

Single trained PINN prices across full (S, t, σ) domain — subsumes an entire B-S grid library.

Correct Physical Behaviour

All three boundary conditions satisfied without explicit labelled data — encoded in the loss function.

Exact Greeks via Autograd

Delta and Gamma computed analytically as a zero-cost by-product. No finite-difference overhead.

Theoretical Guarantee

L^2-error bounded by training loss per De Ryck & Mishra (2022) — dimension-independent bound.

DEPLOYMENT ROADMAP

Production Roadmap: From Research to Infrastructure

1 Phase I — REST Microservice

Deploy trained PINN as API endpoint. Latency: ~0.1ms/CPU, ~0.01ms batched GPU. Throughput: 10,000+ prices/sec.

2 Phase II — Heston Extension

Extend inputs to (S, v, t) and minimise 2D Heston PDE residual. Captures volatility smile — critical for index options.

3 Phase III — Multi-Asset Baskets

d-dimensional Kolmogorov PDE. $O(d^2)$ PINN parameters vs. 100^d FDM grid. Tractable for d=5 to 20.

4 Phase IV — Bayesian PINNs

Posterior distributions over price surface via variational inference. Confidence intervals for σ-uncertainty propagation into risk metrics.

References: De Ryck & Mishra (2022) · Raissi et al. (2019) · Black & Scholes (1973) · Kingma & Ba (2014) · Li et al. (2020) · Yang et al. (2021) · Buehler et al. (2019) · NSE Dataset 2008–2024